

You are Claude CoWork using Fable 5. Build a sleek, modern, secure, mobile-friendly website that allows users to upload their electric bills and receive a practical DIY solar conversion plan based on their real electricity usage.

The core purpose of this website is to help a homeowner understand what it would take to reduce or eliminate grid dependence using solar panels, batteries, inverters, charge controllers, and related equipment. The site should analyze the user's electric bill, estimate their household energy usage, and generate a straightforward DIY solar plan that can be completed all at once or in phases.

Working name for the site: **Bill-to-Solar DIY Planner**

Final name/domain can be changed later, so make the app easy to rebrand.

Important safety framing: this website may provide educational DIY planning, conceptual wiring diagrams, component explanations, and shopping lists, but it must not encourage unsafe electrical work. It must clearly state that final installation, utility interconnection, service-panel work, transfer switches, grounding/bonding, code compliance, permits, and inspections may require a licensed electrician and approval from the local authority having jurisdiction. All recommendations must emphasize NEC-aligned safety, UL-listed components where applicable, proper fusing/breakers, disconnects, wire sizing, grounding, battery safety, ventilation, and manufacturer instructions.

Build the website as a production-ready web application with the following goals:

1. Users can upload electric bills.
2. The app can extract and analyze electricity usage.
3. The app can ask the user to confirm or correct extracted values.
4. The app generates a DIY solar/off-grid plan based on usage.
5. The app provides whole-home and phased/incremental conversion options.
6. The app provides a component list, estimated quantities, budget ranges, and suggested U.S. purchasing sources.
7. The app explains how each component fits together.
8. The app produces safe conceptual wiring and system-layout guidance without giving reckless energized-panel instructions.
9. The app can export the final plan as a printable PDF.

Use a clean, modern technology stack suitable for deployment. Prefer:

- Next.js or React-based frontend
- TypeScript
- Tailwind CSS
- Supabase or another simple backend for auth, storage, and database
- Server-side upload processing
- OCR/PDF parsing pipeline
- Optional AI analysis layer
- Responsive design for desktop and mobile
- Secure file handling
- Clean component architecture
- Environment-variable based API keys

- Clear README and deployment instructions

The user upload flow must support:

- PDF electric bills
- Photos of paper bills
- Screenshots
- JPEG/PNG/HEIC where feasible
- Manual entry fallback if OCR fails
- Future support for CSV or utility usage exports

Create an upload page where the user can drag and drop files, take a photo, or manually enter their usage. After upload, the system should parse the bill and attempt to extract:

- Utility company name
- Customer name, but store/redact this carefully
- Service address or ZIP code
- Billing period
- Number of billing days
- Total kWh used
- Average daily kWh
- Previous usage if shown
- Rate plan if shown
- Total bill amount
- Energy charges
- Delivery charges
- Demand charges if present
- Taxes/fees if present
- Cost per kWh estimate
- Seasonal usage indicators if multiple bills are uploaded
- Any time-of-use information if available
- Peak/off-peak rates if available
- Net metering information if shown
- Existing solar credits if shown

After extraction, show the user a “Review Your Bill Data” screen. Do not blindly trust OCR. Display each extracted field with a confidence score and allow the user to edit every field before generating the plan. If the bill is unreadable or missing usage data, ask for manual kWh, billing days, ZIP code, and average monthly electric bill.

The plan generator must use the confirmed bill data to calculate practical solar planning estimates. Use clear formulas and show assumptions. At minimum calculate:

- Average daily kWh
- Estimated monthly kWh
- Estimated annual kWh
- Estimated peak household load, using conservative assumptions unless user provides actual peak load

- Solar array size needed in kW
- Number of solar panels needed based on selectable panel wattage
- Battery storage needed in kWh
- Battery bank sizing based on battery chemistry
- Usable battery capacity based on depth of discharge
- Inverter size recommendation
- Charge controller or hybrid inverter requirements
- Estimated space required for panels
- Estimated budget range
- Estimated phase-by-phase implementation cost
- Estimated percent of usage covered by each phase

Use this general planning logic:

- Daily kWh = monthly kWh divided by billing days or average month length
- Solar array size = daily kWh divided by effective sun hours and derating factor
- Use derating assumptions for real-world conditions such as shade, panel angle, inverter loss, temperature, and wiring loss
- Battery storage = daily critical or whole-home load multiplied by desired autonomy days, divided by usable depth of discharge and efficiency
- Inverter size = estimated peak load plus surge margin
- Battery chemistry affects usable capacity, cycle life, ventilation, safety, and replacement cost
- The app should explain that exact design requires site-specific solar access, roof condition, local code, and professional review

The user should be able to choose from multiple plan types:

1. **Whole-Home Solar Conversion**

2. Designed to offset most or all household electricity usage
3. Includes panels, inverter, battery storage, disconnects, monitoring, and optional generator backup
4. Clearly explains that whole-home off-grid conversion is expensive and must be designed carefully

5. **Critical Loads Backup Plan**

6. Designed to power essentials first
7. Include refrigerator/freezer, lights, modem/router, phone charging, small appliances, medical devices if specified, well pump if specified, HVAC only if realistic
8. Recommend a critical loads panel or transfer switch conceptually
9. Make clear that panel work requires proper permitting and qualified help

10. **Phased DIY Solar Plan**

11. Phase 1: reduce load and identify critical circuits

12. Phase 2: small solar + battery setup for essentials
13. Phase 3: expand battery bank and solar array
14. Phase 4: add hybrid inverter, critical loads panel, or whole-home system
15. Phase 5: optional off-grid transition or grid-tied/hybrid setup

16. Budget Starter Plan

17. Smallest realistic setup for meaningful benefit
18. Could include portable panels, charge controller, battery, inverter, and basic emergency loads
19. Must avoid implying that a tiny kit can run an entire house

20. Off-Grid Cabin / Detached Building Plan

21. For sheds, cabins, workshops, RV-style setups, or small independent loads
22. Lower-voltage DC guidance can be more DIY-oriented, but still include fuses, wire sizing, grounding, and safety

The generated plan should include these sections:

- Summary of electric bill analysis
- Estimated household usage
- Recommended solar strategy
- Whole-home option
- Phased option
- Critical-loads option
- Component list
- Budget shopping list
- Battery comparison
- Solar panel sizing
- Inverter sizing
- Charge controller or hybrid inverter sizing
- Mounting considerations
- Safety considerations
- Conceptual wiring/system diagram
- Installation sequence
- What a DIYer can reasonably do
- What should be handled by a licensed electrician
- Permits, inspections, and utility approval checklist
- Maintenance checklist
- Expansion roadmap
- Final assumptions and limitations

Component recommendations must include the following categories:

- Solar panels
- Roof mount or ground mount racking
- Combiner box, if needed
- MPPT charge controller or hybrid inverter
- Inverter or inverter/charger
- Battery bank
- Battery rack or enclosure
- Battery management considerations
- DC breakers/fuses
- AC breakers where appropriate
- Disconnect switches
- Transfer switch or critical loads panel concept
- Surge protection
- Grounding and bonding components
- Properly sized wire/cable
- Lugs, busbars, conduit, glands, labels
- Monitoring system
- Safety labels
- Fire safety considerations
- PPE
- Optional backup generator input
- Optional EV charging considerations if user has high usage

Battery options should include clear comparison between:

- LiFePO4 batteries
- Flooded lead-acid batteries
- AGM batteries
- Gel batteries
- Marine/deep-cycle batteries

The app should explain that marine batteries can be useful for budget/emergency starter systems but are usually not ideal for serious whole-home solar due to cycle life and usable capacity limits. The app should not mix battery chemistries in one bank. It should explain ventilation needs for flooded lead-acid batteries, proper charging profiles, temperature considerations, fusing, and enclosure requirements.

The website must include inexpensive U.S. sourcing guidance. Create a vendor/source system that can be updated over time. Include categories such as:

- Big box hardware stores
- Solar-specific retailers
- Battery suppliers
- Online marketplaces
- Electrical supply houses
- Local electricians/solar installers for code-critical work

Suggested source examples to include as editable vendor entries:

- Home Depot
- Lowe's
- Harbor Freight
- Amazon
- Walmart
- Sam's Club
- Costco
- Renogy
- Eco-Worthy
- Signature Solar
- Current Connected
- The AltE Store
- Northern Arizona Wind & Sun
- Local electrical supply stores
- Local battery suppliers
- Local solar installers for inspections, interconnection, and service panel work

Do not hardcode current prices as permanent facts. Use estimated budget ranges and make the system easy to update. The app should say "verify current pricing, shipping, warranty, and compatibility before purchasing."

The website should have the following pages:

1. Home Page

2. Clear headline: "Upload your electric bill. Get a DIY solar roadmap."
3. Short explanation of what the app does
4. Upload button
5. Safety disclaimer
6. Examples of plans: starter, critical loads, phased, whole home

7. Upload Bill Page

8. Drag-and-drop upload
9. Camera/photo upload
10. PDF upload
11. Manual entry fallback
12. Privacy note

13. Bill Review Page

14. Show extracted data
15. Confidence indicators
16. Editable fields
17. "Confirm and Generate Plan" button
- 18. Solar Plan Page**
19. Usage summary
20. Recommended plan type
21. Whole-home option
22. Phased option
23. Component list
24. Budget ranges
25. Conceptual system diagram
26. Safety checklist
27. Export PDF button
- 28. Component Library**
29. Educational explanations of panels, batteries, inverters, controllers, transfer switches, disconnects, breakers, wire, grounding, and monitoring
30. Plain-English descriptions
31. "What this does"
32. "Why you need it"
33. "Common mistakes"
34. "DIY difficulty level"
35. "When to call an electrician"
- 36. Battery Comparison Page**
37. Compare LiFePO₄, lead-acid, AGM, gel, and marine batteries
38. Show pros, cons, approximate usable capacity, lifespan, maintenance, and safety considerations
- 39. Solar Basics Page**
40. Explain watts, kilowatts, kWh, voltage, amperage, inverter surge, battery depth of discharge, sun hours, and derating
41. Keep it simple and practical

42. **Safety & Code Page**

43. Clear warnings

44. NEC/local code reminders

45. Permitting checklist

46. Utility interconnection checklist

47. Electrician-required tasks

48. Battery safety

49. Fire safety

50. Roof safety

51. Working around service panels

52. **Saved Plans / Dashboard**

53. Optional account-based plan storage

54. Let users save multiple bills and compare plans over time

The app should include privacy and security features:

- Do not expose uploaded bills publicly
- Store files securely
- Allow users to delete uploaded bills
- Redact or avoid storing sensitive personal information where possible
- Encrypt or protect stored documents according to best practices
- Explain what data is used and why
- Allow guest mode if feasible, but recommend account creation for saved plans
- Use signed URLs for private files
- Validate file type and size
- Prevent malicious uploads
- Sanitize extracted text
- Use secure server-side processing

The app should include a bill parsing/OCR pipeline:

- Extract text from PDFs
- Run OCR on images
- Normalize utility bill language
- Detect kWh and billing period
- Use confidence scoring
- Provide a manual correction interface
- Store the extracted bill data separately from the original file
- Allow multiple bills to improve seasonal accuracy
- Show warnings when one bill may not represent annual usage

If AI is used to interpret the bill, enforce a structured JSON output. The AI should return:


```

{
  "utilityCompany": "",
  "billingStartDate": "",
  "billingEndDate": "",
  "billingDays": null,
  "totalKwh": null,
  "totalBillAmount": null,
  "estimatedCostPerKwh": null,
  "serviceZip": "",
  "ratePlan": "",
  "demandCharge": null,
  "timeOfUse": false,
  "confidence": {
    "utilityCompany": 0,
    "billingDays": 0,
    "totalKwh": 0,
    "totalBillAmount": 0,
    "serviceZip": 0
  },
  "needsUserReview": true,
  "notes": []
}

```

The generated solar plan should also be structured. Use a plan object like:

```

{
  "usageSummary": {
    "averageDailyKwh": 0,
    "estimatedMonthlyKwh": 0,
    "estimatedAnnualKwh": 0,
    "estimatedCostPerKwh": 0
  },
  "recommendedStrategy": "",
  "wholeHomePlan": {
    "solarArrayKw": 0,
    "panelWattage": 0,
    "estimatedPanelCount": 0,
    "batteryKwhRecommended": 0,
    "inverterKwRecommended": 0,
    "estimatedBudgetLow": 0,
    "estimatedBudgetHigh": 0
  },
  "criticalLoadsPlan": {
    "dailyCriticalKwh": 0,
    "solarArrayKw": 0,
    "batteryKwhRecommended": 0,

```

```

    "estimatedBudgetLow": 0,
    "estimatedBudgetHigh": 0
  },
  "phases": [
    {
      "phaseNumber": 1,
      "name": "",
      "goal": "",
      "components": [],
      "estimatedCostLow": 0,
      "estimatedCostHigh": 0,
      "estimatedUsageOffsetPercent": 0,
      "notes": []
    }
  ],
  "componentList": [],
  "safetyWarnings": [],
  "electricianRequiredTasks": [],
  "assumptions": []
}

```

Create a strong UI experience:

- Modern, clean, trustworthy design
- Solar/energy-inspired visual style
- Use warm yellows, deep navy/charcoal, white, and subtle green accents
- Avoid looking gimmicky
- Make the app feel practical and reliable
- Use cards, progress steps, icons, and simple charts
- Show bill usage visually
- Show phase comparison visually
- Show cost range comparison
- Use plain language for non-technical homeowners
- Make everything mobile responsive

The app should include a conceptual wiring/system layout section. This should show safe, educational, non-dangerous diagrams such as:

Solar panels → combiner/disconnect → charge controller or hybrid inverter → battery bank → inverter → critical loads panel / selected loads

For whole-home or grid-connected systems, clearly state:

- Utility interconnection must follow local utility requirements
- Service panel tie-ins should be performed by qualified professionals
- Transfer switches must be properly rated and installed
- Backfeeding a home without proper transfer equipment is dangerous and must not be done

- Grounding/bonding must be code-compliant
- Wire sizing and overcurrent protection must be calculated for the exact equipment and installation

Do not create instructions that tell an unqualified user to open an energized main panel or bypass the utility meter. Instead, provide a checklist of what to discuss with an electrician.

Include an “Installation Difficulty” rating for each phase:

- Beginner: planning, bill review, load inventory, small portable solar kit setup
- Intermediate: low-voltage DC assembly with proper fusing and manufacturer instructions
- Advanced: larger battery/inverter systems, mounting, conduit planning
- Licensed professional: service panel, transfer switch, utility interconnection, roof structural issues, final inspection

The app should include load inventory functionality. Let the user optionally enter appliances and loads:

- Refrigerator
- Freezer
- Lights
- Well pump
- HVAC
- Window AC
- Heat pump
- Electric water heater
- Electric stove
- Microwave
- Washer/dryer
- Internet/router
- Medical equipment
- EV charger
- Power tools
- Other custom loads

For each load, allow estimated watts and hours per day. Use this to refine the critical loads plan.

The app should warn users that electric heat, central AC, electric water heaters, dryers, ovens, and EV chargers drastically increase solar/battery requirements.

The app should support the following plan output styles:

- Concise homeowner summary
- Detailed technical estimate
- Printable shopping list
- Electrician discussion checklist
- Step-by-step phased roadmap

Include PDF export with:

- User's confirmed usage summary
- Recommended plan
- Phases
- Component list
- Budget range
- Safety checklist
- Questions for electrician
- Assumptions and limitations

Build the project with a clean database structure. Suggested tables:

- users
- uploaded_bills
- extracted_bill_data
- solar_plans
- plan_phases
- component_library
- vendor_sources
- saved_load_profiles
- plan_exports

Each uploaded bill should connect to extracted data and generated plans.

Create sample/mock data so the app works even before live OCR or AI APIs are fully configured. Include a demo bill analysis mode with fake but realistic values so the UI can be tested.

The app should include clear error states:

- Upload failed
- Unsupported file type
- OCR could not read bill
- Missing kWh usage
- User must confirm values
- AI parsing unavailable
- Plan generation unavailable
- PDF export failed

The app must never hallucinate precision. If data is estimated, label it as estimated. If the app does not know the roof size, panel orientation, shade, or local code, clearly state those assumptions.

The app should be built so future integrations are easy, including:

- NREL PVWatts or similar solar production estimates
- DSIRE or similar incentive lookup
- Utility rate APIs
- Live component price lookups

- Vendor affiliate links
- Google/Facebook login
- Stripe payment for premium plans
- Contractor/electrician referral marketplace
- AI chat assistant for solar planning
- User dashboard with multiple properties

Initial MVP requirements:

- Home page
- Upload page
- Manual bill entry fallback
- OCR/PDF text extraction placeholder or implementation
- Bill review/edit screen
- Solar plan generation
- Whole-home, critical-loads, and phased options
- Component shopping list
- Battery comparison
- Conceptual wiring/system diagram
- Safety/code page
- PDF export
- Saved plan dashboard if auth is included
- Clean README with setup and deployment instructions

Acceptance criteria:

- A user can upload or manually enter electric bill data.
- The app extracts or accepts kWh usage and billing period.
- The user can review and correct bill data.
- The app calculates average daily usage.
- The app generates at least three plan options: whole-home, critical loads, and phased.
- The app generates a realistic component list.
- The app provides budget ranges and suggested U.S. sources.
- The app includes safety warnings and electrician-required tasks.
- The app includes a conceptual system diagram.
- The app can export the plan to PDF.
- The UI is clean, modern, responsive, and easy to understand.
- The code is organized and documented.
- Sensitive uploads are handled securely.

Create the complete application using Table 5. Make reasonable implementation decisions where needed. Do not stop to ask non-blocking questions. Prioritize a functional MVP with clean architecture that can be expanded later.